

BAULKHAM HILLS HIGH SCHOOL

HIGHER SCHOOL CERTIFICATE

TRIAL EXAMINATION

2009

MATHEMATICS

EXTENSION 1

Time Allowed - Two hours
(Plus five minutes reading time)

General Instructions

- Attempt ALL questions
- Start each question on a new page
- All necessary working should be shown
- Write your student number at the top of each page of answer sheets
- Board approved calculators may be used
- Write using black or blue pen

Question 1

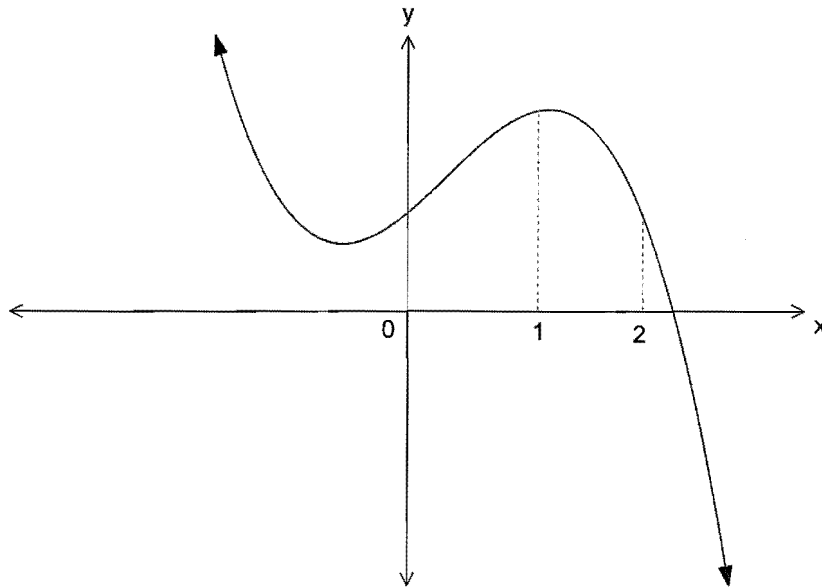
- a) Evaluate $\sum_{n=0}^4 (1-2n)$ 1
- b) Solve $\frac{x}{x-2} \geq 2$ 3
- c) Find the coordinates of the point P which divides the interval AB **externally** in the ratio 1 : 3, given A= (1,4) and B= (5,2) 2
- d) Evaluate
- i) $\int_1^2 \frac{1}{\sqrt{4-x^2}} dx$ 2
- ii) $\int_{-1}^0 x\sqrt{1+x} dx$, using the substitution $u = 1+x$ 4

Question 2

- a) Simplify $\frac{{}^nC_2}{{}^nC_1}$ 1
- b) A circular oil slick is spreading over a bay, such that its radius is increasing at a constant rate of 0.1 m/s
- What is the radius when the area is increasing at $2\pi \text{ m}^2/\text{s}$? 3
- c) Simplify $\sin 2\theta (\tan \theta + \cot \theta)$ 2
- d) Consider the function $f(x) = 3 \cos^{-1} \left(\frac{x}{2} \right)$
- i) Sketch the graph $y = f(x)$ 3
- ii) Find the gradient of the tangent to the curve at the point on it where $x = \sqrt{3}$ 2

Question 3

- a) The polynomial $y = x^2 + 2x + 2 - x^3$ has only one root, as shown on the diagram below



Using one application of Newton's method and $x=2$ as the first approximation, find a better approximation this root.

3

- b) Solve for $0 \leq \theta \leq 2\pi$: $\cos 2\theta = \cos \theta$

3

- c) Find the term independent of x in the expression of $\left(x - \frac{1}{2x^3}\right)^{20}$

3

- d) A particle is moving with acceleration $\ddot{x} = -9x$ and is initially stationary at $x = 4$

- i) Find v^2 as a function of x

2

- ii) What is the particle's maximum speed?

1

Question 4

- a) Find $\int \cos^2 2x \, dx$

3

- b) Prove, by Mathematical Induction that

4

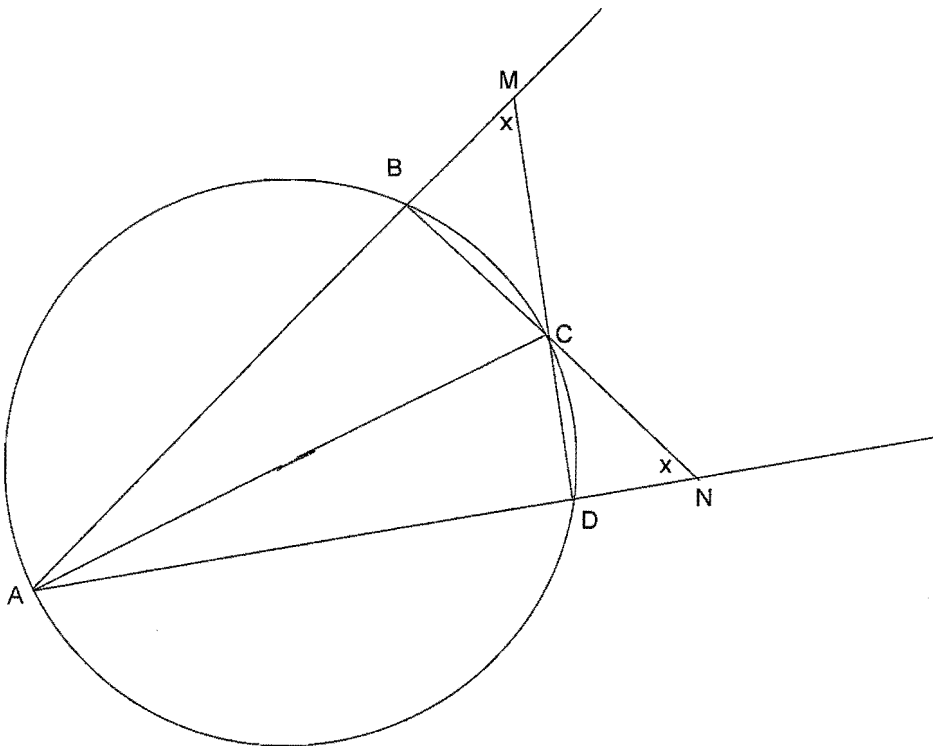
$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + n(n+1)^2 = \frac{1}{12} n(n+1)(n+2)(3n+5)$$

for all positive integers n

- c) i) Express $\cos \theta - \sqrt{3} \sin \theta$ in the form $R \cos (\theta + \alpha)$ where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$ 2
- ii) Solve for $0 \leq \theta \leq 2\pi$, $\cos \theta - \sqrt{3} \sin \theta = 1$ 2
- iii) What is the maximum value of $\cos \theta - \sqrt{3} \sin \theta$? 1

Question 5

- a) If $\alpha = \sin^{-1} \left(\frac{8}{17} \right)$ and $\beta = \tan^{-1} \left(\frac{3}{4} \right)$ calculate the exact value of $\sin (\alpha - \beta)$ 3
- b) Find the greatest coefficient in the expansion of $(5 + 2x)^9$ 4
- c)

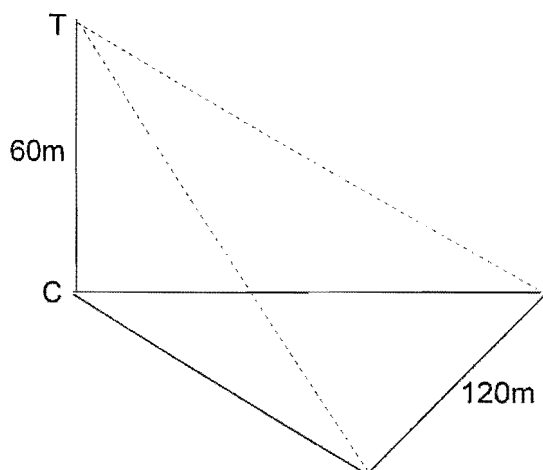


In the figure, ABM , BCN and ADN are straight lines and $\angle AMD = \angle BNA = x$

- i) Copy the diagram and prove that $\angle ABC = \angle ADC$ 3
- ii) Hence, prove that AC is a diameter. 2

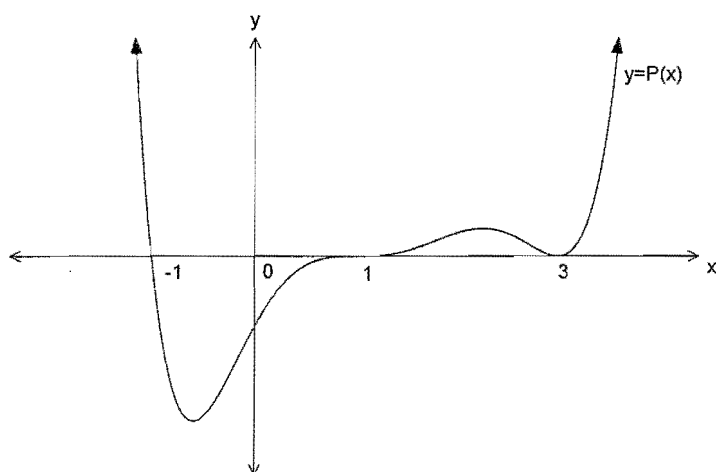
Question 6

- a) The angles of elevation of the top of a tower TC , 60m high are measured from two points, A and B , which are 120m apart. (A , B and C are all on level ground). These angles of elevation are found to be 30° from A and 53° from B . 5

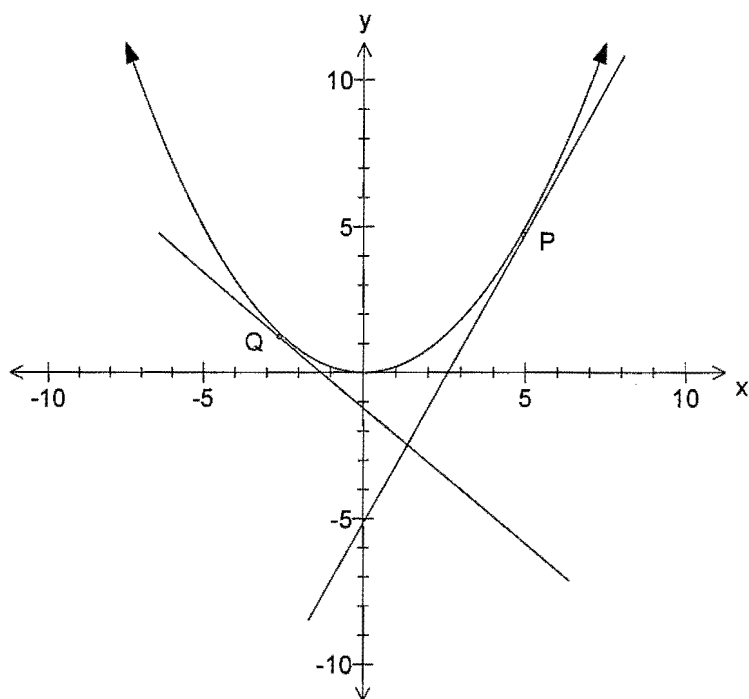


If A bears $038^\circ T$ from the foot of the tower, find the possible bearings of B from the tower. Answer correct to nearest degree. (Copy the diagram first)

- b) Write down a possible equation $y = P(x)$ for the polynomial function sketched below. 3



- c) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$



The tangents at P and Q intersect at 45°

- | | | |
|------|---|----------|
| i) | Show that the gradient of the tangent at P is p | 1 |
| ii) | Show that $ p - q = 1 + pq $ | 1 |
| iii) | If $p = 2$, evaluate q | 2 |

Question 7

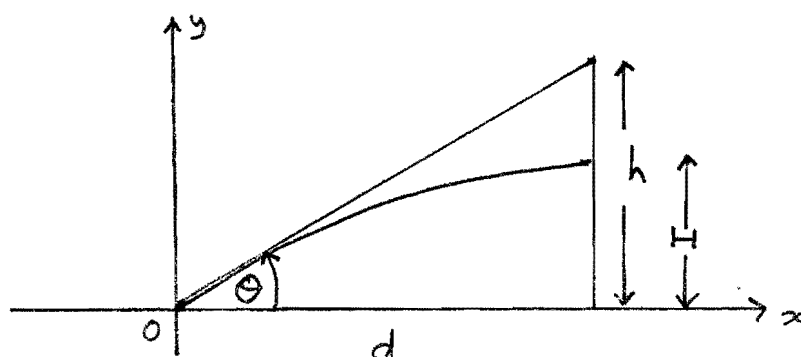
- a) The roots of $x^3 + kx^2 - 54x - 216 = 0$ form a geometric progression. Find the roots. 4

- b) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ 1

- c) A target is hung on a wall at a height of h metres.

A small cannon, which fires a lead slug, is located on the floor, d metres from the wall. The initial velocity, V , at which the slug is fired is adjustable.

The cannon is aimed at the bullseye on the target at an angle of elevation of θ degrees. At the instant the cannon is fired, the target is released and falls vertically downwards under the force of gravity, g



Given that $\ddot{x} = 0$ and $\ddot{y} = -g$:

- i) Show that the position of the lead slug at time t is given by 2

$$x = Vt \cos \theta \text{ and } y = -\frac{gt^2}{2} + Vt \sin \theta$$

- ii) Show that the slug hits the wall at a vertical height of 2

$$H = -\frac{g d^2 \sec^2 \theta}{2V^2} + d \tan \theta$$

- iii) Experiments with the cannon show that the slug always hits the bullseye, regardless of the initial velocity. Explain why this is always so. 3

End of examination

SOLUTIONS.

Q1.

a) $1 + (-1) + (-3) + (-5) + (-7)$

$= -15$

(1)

b) $\frac{x^{1(x-2)^2}}{x^2} \geq \frac{2(x-2)^2}{x^2}$ (1)

(NOTE: $x \neq 2$)

$x^2 - 2x \geq 2x^2 - 8x + 8$

$0 \geq x^2 - 6x + 8$

$0 \geq (x-4)(x-2)$

$(x-4)(x-2) \leq 0$

Roots: $x=4, 2$

Soln:

$2 < x \leq 4$

(1)

c) $A(1, 4)$ $B(5, 2)$

$1:3$

$x = \frac{1(5) - 3(1)}{1-3}, y = \frac{1(2) - 3(4)}{1-3}$

$= \frac{2}{-2}$

$= -1$

$= \frac{-10}{-2}$

$= 5$

$\therefore P = (-1, 5)$ (1) x-value
(1) y-value

d) i) $\left[\sin^{-1} \frac{x}{2} \right]_1^2$ (1)

$= \sin^{-1} 1 - \sin^{-1} \frac{1}{2}$

$= \frac{\pi}{2} - \frac{\pi}{6}$

$= \frac{\pi}{3}$ (1)

ii) $\int_{-1}^0 x \sqrt{1+x} \cdot dx$

$u = 1+x \quad x = u-1$

$du = dx$

If $x = -1, u = 0$

If $x = 0, u = 1$

$= \int_0^1 (u-1) \cdot u^{1/2} \cdot du$ (1)

$= \int_0^1 u^{3/2} - u^{1/2} \cdot du$

$= \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_0^1$ (1)

$= \left(\frac{2}{5} - \frac{2}{3} \right) - (0 - 0)$

$= -\frac{4}{15}$ (1)

Q2. [11 marks]

a) $\frac{n!}{(n-2)! 2!} \div \frac{n!}{1! (n-1)!}$

$= \frac{n!}{2! (n-2)!} \times \frac{(n-1)!}{n!}$

$= \frac{n-1}{2!}$

$= \frac{n-1}{2}$ (1)

b) $A = \pi r^2 \quad \frac{dA}{dr} = 2\pi r$

$\frac{dA}{dt} = \frac{dA}{dr} \cdot \frac{dr}{dt}$ (1)

$2\pi = 2\pi r \times 0.1$ (1)

$2\pi = 0.2\pi r$

$r = 10 \text{ m}$ (1)

c) $2 \sin \theta \cos \theta \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$

$= 2 \sin^2 \theta + 2 \cos^2 \theta$ (1)

$= 2(\sin^2 \theta + \cos^2 \theta)$

$= 2 \times 1$

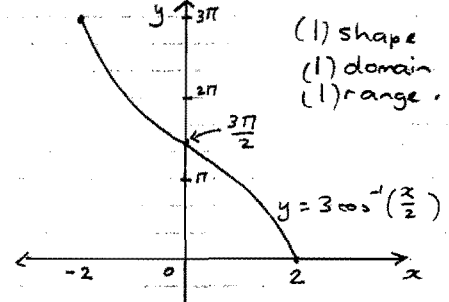
$= 2$ (1)

d) $f(x) = 3 \cos^{-1} \left(\frac{x}{2} \right)$

(i) D: $-1 \leq \frac{x}{2} \leq 1$

$-2 \leq x \leq 2$

R: $0 \leq f(x) \leq 3\pi$



(ii) $\frac{dy}{dx} = 3 \cdot \frac{-1}{\sqrt{4-x^2}}$ (1)

$= \frac{-3}{\sqrt{4-x^2}}$

$= -3$ (1)

Q3.

$$a) a_1 = a_0 - \frac{f(a_0)}{f'(a_0)} \quad \begin{cases} a_0 = 2 \\ f(2) = 4 + 4 + 2 - 8 = 2 \\ f'(x) = 2x + 2 - 3x^2 \\ f'(2) = 4 + 2 - 12 = -6 \end{cases}$$

$$= 2 - \frac{2}{-6} \quad (1)$$

$$= \frac{2}{3} \quad (1)$$

$$b) 2 \cos^2 \theta - 1 = \cos \theta$$

$$\left. \begin{aligned} 2 \cos^2 \theta - \cos \theta - 1 &= 0 \\ (2 \cos \theta + 1)(\cos \theta - 1) &= 0 \end{aligned} \right\} \quad (1)$$

$$\cos \theta = -\frac{1}{2} \quad \cos \theta = 1$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 0, 2\pi$$

$$c) T_{k+1} = {}^{20}C_k \cdot x^{20-k} \cdot (2x^3)^{-k} \cdot (-1)^k \quad (1)$$

$$= {}^{20}C_k \cdot x^{20-k} \cdot 2^{-k} \cdot x^{-3k} \cdot (-1)^k$$

$$= {}^{20}C_k \cdot x^{20-4k} \cdot 2^{-k}$$

$$\text{Indep. of } x \text{ when } 20-4k=0 \text{ i.e. } k=5 \quad (1)$$

Term indep. of x is:

$$T_6 = {}^{20}C_5 \cdot x^0 \cdot 2^{-5} = {}^{20}C_5 \cdot 2^{-5} \quad (1)$$

(or -484.5)

$$d) (i) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$$

$$\left. \begin{aligned} t=0 \\ x=4 \\ v=0 \end{aligned} \right\}$$

$$v^2 = 2 \int -9x \cdot dx$$

$$v^2 = -9x^2 + c$$

$$0 = -9(16) + c$$

$$c = 144$$

$$v^2 = -9x^2 + 144$$

$$(ii) \text{ Max speed when } \ddot{x} = 0$$

$$\text{when } x=0$$

$$\text{Max } v^2 = 144$$

$$\therefore \text{Max } |v| = 12 \quad (1)$$

Q4.

$$a) \int \cos^2 2x \cdot dx$$

$$= \int \frac{1}{2} (1 + \cos 4x) \cdot dx \quad (1)$$

$$= \frac{1}{2} \left(x + \frac{1}{4} \sin 4x \right) + c \quad (2)$$

$$b) \text{ If } n=1 \quad \text{LHS} = 1 \times 2^2 = 4$$

$$\text{RHS} = \frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \cdot 8 = 4$$

$$\therefore \text{LHS} = \text{RHS}, \text{ so true for } n=1 \quad (1) \text{ Prove for } n=1$$

Assume true for $n=k$

i.e. Assume

$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 = \frac{1}{12} k(k+1)(k+2)(3k+5)$$

Need to prove true for $n=k+1$

i.e. Prove

$$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots + k(k+1)^2 + (k+1)(k+2)^2$$

$$= \frac{1}{12} (k+1)(k+2)(k+3)(3k+8)$$

$$\text{LHS} = \frac{1}{12} k(k+1)(k+2)(3k+5) + (k+1)(k+2)^2$$

by assumption

$$= \frac{1}{12} (k+1)(k+2) \left(k(3k+5) + 12(k+2) \right) \quad (1)$$

$$= \frac{1}{12} (k+1)(k+2) (3k^2 + 17k + 24) \quad (1)$$

$$= \frac{1}{12} (k+1)(k+2)(k+3)(3k+8)$$

$$= \text{RHS}$$

∴ If true for $n=k$, then also true for $n=k+1$.

Now statement is true for $n=1$.

∴ Also true for $n=2, 3, 4, \dots$

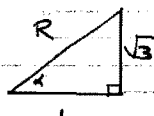
By induction, true for all positive integers n .

[+1 if conclusion inappropriate]

c) (i) $R \cos(\theta + \alpha) = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$

$R \cos \alpha = 1/R$

$R \sin \alpha = \sqrt{3}/R$



$\alpha = \frac{\pi}{3}, R = 2$

$\alpha \leftarrow (1)$
 $R \leftarrow (1)$

∴ $\cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + \frac{\pi}{3})$

(ii) $2 \cos(\theta + \frac{\pi}{3}) = 1$

$0 \leq \theta \leq 2\pi$

$\cos(\theta + \frac{\pi}{3}) = \frac{1}{2}$

$\frac{\pi}{3} \leq \theta + \frac{\pi}{3} \leq \frac{7\pi}{3}$

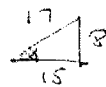
$\theta + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$

$\theta = 0, \frac{4\pi}{3}, 2\pi$ (2)

(iii) 2 (1)

Q5.

a) $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ (1)



$= \frac{8}{17} \cdot \frac{4}{5} - \frac{15}{17} \cdot \frac{3}{5}$ (2)



$= \frac{-13}{85}$

-1 per error

b) T_{k+1}

b) Ratio of terms, $\frac{T_{k+1}}{T_k} = \frac{n-k+1}{k} \cdot \frac{b}{a}$

$= \frac{10-k}{k} \cdot \frac{2x}{5}$

Ratio of coeffs: $\frac{10-k}{k} \cdot \frac{2}{5} > 1$ for coeffs. (1) increasing

$20 - 2k > 5k$

$20 > 7k$ (1)

$k < 2\frac{6}{7}$

$k = 2 \dots$ greatest such integer (1)

Max coeff. (is in T_3) = ~~20~~ $2^2 \cdot 5^7 \cdot 2^2$ (1)

$= 11250000$

c) (i) $\angle BCM = \angle DCN$ (vertically opp. $\angle$ s equal) (1)

$\angle ABC = x + \angle BCM$
 $\angle ADC = x + \angle DCN$ } ext. $\angle$ of Δ
= sum of int. opp. $\angle$ s (1)

∴ $\angle ABC = \angle ADC$ (addition of equals). (1)

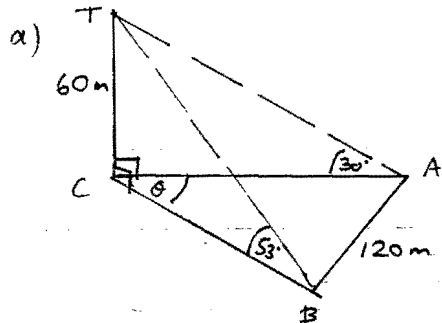
(ii) $\angle ABC + \angle ADC = 180$ (opp. $\angle$ s of cyclic quad. supplementary) (1)

but $\angle ABC = \angle ADC$ from (i)

∴ $\angle ABC = \angle ADC = 90^\circ$ (1)

∴ AC is a diameter ($\angle$ in semicircle is 90°).

Q6.



$$\left. \begin{aligned} \tan 30^\circ &= \frac{60}{AC} & AC &= \frac{60}{\tan 30^\circ} \div 103.92 \\ \tan 53^\circ &= \frac{60}{BC} & BC &= \frac{60}{\tan 53^\circ} \div 45.21 \end{aligned} \right\} (1)$$

$$\begin{aligned} \cos \theta &= \frac{AC^2 + BC^2 - AB^2}{2 \cdot AC \cdot BC} \\ &= \frac{103.92^2 + 45.21^2 - 120^2}{2 \times 103.92 \times 45.21} \\ &= -0.1657 \end{aligned}$$

$$\theta = 100^\circ \quad (\text{nearest } ^\circ).$$

B may be 100° clockwise from A
or 100° anticlockwise from A

$$\therefore \text{Bearing of B} = \underline{138^\circ \text{T or } 298^\circ \text{T}} \quad (1) \quad \text{both bearings}$$

b) $y = (x+1)(x-1)^3(x-3)^2$

c) (i) $y = \frac{x^2}{4a} \quad \frac{dy}{dx} = \frac{2x}{4a} = \frac{2(2ap)}{4a} \quad \text{at P.}$
 $= p. \quad (1)$

(ii) At P : $m_1 = p$ from above

At Q : $m_2 = q$ similarly

$$\therefore \tan 45^\circ = \left| \frac{m_1 - m_2}{1} \right| \quad \text{in formula}$$

$$1 = \left| \frac{p-q}{1+pq} \right| \quad (1) \quad (\text{must show } \tan 45^\circ).$$

$$|1+pq| = |p-q|$$

(iii) $|1+2q| = |2-q|$

$$1+2q = 2-q \quad \text{or} \quad 1+2q = -2+q$$

$$3q = 1$$

$$\underline{q = \frac{1}{3}} \quad (1)$$

$$\underline{q = -3} \quad (1)$$

Q7.

a) Let roots $= \frac{\alpha}{r}, \alpha, \alpha r$

$$\text{Sum} \quad \frac{\alpha}{r} + \alpha + \alpha r = -k$$

$$\text{Prod.} \quad \alpha^3 = 216 \quad (1)$$

$$\alpha = 6$$

Since $\alpha = 6$ is a root,
 $216 + 36k - 324 - 216 = 0$
 $36k = 324$

$$k = 9 \quad (1)$$

$$\therefore \frac{6}{r} + 6 + 6r = -9$$

$$6 + 6r + 6r^2 = -9r$$

$$6r^2 + 15r + 6 = 0$$

$$2r^2 + 5r + 2 = 0$$

$$(2r+1)(r+2) = 0$$

$$r = -\frac{1}{2} \quad \text{or} \quad -2 \quad (1)$$

$\therefore$ Roots $\frac{6}{r}, 6, 6r$

$$= -3, 6, -12 \quad (1)$$

b) $\frac{3}{5}$ (1)

c) (i)

$$\begin{aligned} \ddot{x} &= 0 & \ddot{y} &= -g \\ \dot{x} &= \int 0 \cdot dt & \dot{y} &= \int -g \cdot dt \\ &= c & &= -gt + c \\ t=0 \left. \begin{array}{l} \dot{x} = V \cos \theta \\ \dot{y} = V \sin \theta \end{array} \right\} & & t=0 \left. \begin{array}{l} \dot{y} = V \sin \theta \end{array} \right\} \\ \therefore \dot{x} &= V \cos \theta & & V \sin \theta = 0 + c \\ x &= \int V \cos \theta \cdot dt & \dot{y} &= -gt + V \sin \theta \\ &= (V \cos \theta)t + c & y &= \int -gt + V \sin \theta \cdot dt \\ t=0 \left. \begin{array}{l} x=0 \end{array} \right\} & 0 = 0 + c & & = -\frac{1}{2}gt^2 + (V \sin \theta)t + c \\ c &= 0 & & t=0 \left. \begin{array}{l} y=0 \end{array} \right\} 0 = 0 + 0 + c \\ \underline{x = Vt \cos \theta} & & \underline{y = -\frac{1}{2}gt^2 + V \sin \theta \cdot t} & (1) \end{aligned}$$

(ii) The slug hits the wall when $\begin{cases} x=d \\ y=H \end{cases}$

$$d = Vt \cos \theta$$

$$t = \frac{d}{V \cos \theta} \quad \text{--- (1)}$$

Then

$$\begin{aligned} H &= -\frac{1}{2}g \cdot \left(\frac{d}{V \cos \theta}\right)^2 + \cancel{V} \cdot \frac{d}{V \cos \theta} \cdot \sin \theta \\ &= -\frac{1}{2} \cdot g \cdot \frac{d^2}{V^2 \cos^2 \theta} + d \cdot \tan \theta \\ &= \frac{-gd^2}{2V^2 \cos^2 \theta} + d \tan \theta \quad (1) \end{aligned}$$

$$\therefore H = \frac{-gd^2}{2V^2} \cdot \sec^2 \theta + d \tan \theta$$

(iii) Target falls vertically from rest

Initial: $\begin{aligned} t &= 0 \\ y &= h \\ \dot{y} &= 0 \\ \ddot{y} &= -g \end{aligned}$

$$\ddot{y} = -g$$

$$\dot{y} = -gt + c$$

$$t=0 \left. \begin{array}{l} \dot{y}=0 \end{array} \right\} \begin{aligned} 0 &= 0 + c \\ c &= 0 \end{aligned}$$

$$\dot{y} = -gt$$

$$y = -\frac{1}{2}gt^2 + c$$

$$t=0 \left. \begin{array}{l} y=h \end{array} \right\} \begin{aligned} h &= 0 + c \\ c &= h \end{aligned}$$

$$y = -\frac{1}{2}gt^2 + h \quad \leftarrow \text{height of target. (1)}$$

$$(\text{or } y = h - \frac{1}{2}gt^2)$$

When the slug hits the wall, $t = \frac{d}{V \cos \theta}$

At this time, $y = -\frac{1}{2}g \cdot \left(\frac{d}{V \cos \theta}\right)^2 + h$

$$= -\frac{gd^2}{2V^2 \cos^2 \theta} + h$$

$$= \frac{-gd^2}{2V^2} \cdot \sec^2 \theta + h \quad (1)$$

but $\tan \theta = \frac{h}{d}$

$$h = d \tan \theta$$

$$y = \frac{-gd^2}{2V^2} \cdot \sec^2 \theta + d \tan \theta \quad (1)$$

i.e. $y = H$

Height of
target

Height of
slug

$\therefore$ It will always
hit.